



What Your Python Code Sounds Like: Debugging Through Sound

~Andrea Mirarchi – PyCon Italia 2026

```
Traceback (most recent call last):  
  File "main.py", line 12, in <mod>  
    print(10 / 0)  
ZeroDivisionError: division by zero
```

```
Traceback (most recent call last):  
  File "main.py", line 8, in <mod>  
    int("ciao")  
ValueError: invalid literal...
```

```
Traceback (most recent call last):  
  File "main.py", line 15, in <mod>  
    print(lista[5])  
IndexError: list index out of range
```

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File "main.py", line 12, in <mod>

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Console Triggers



Console Triggers

ZeroDivisionError

ValueError

IndexError

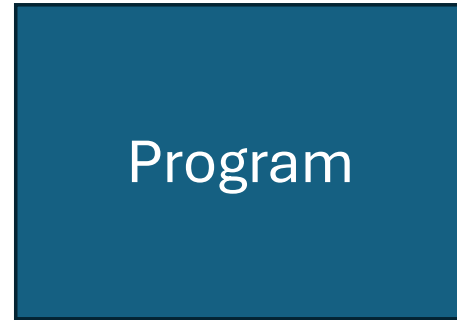
KeyError

TypeError

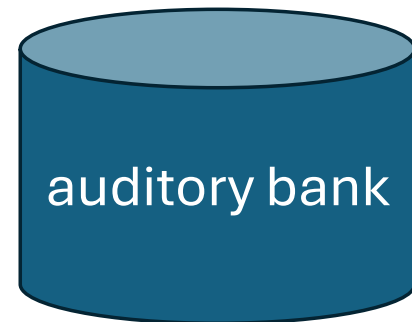
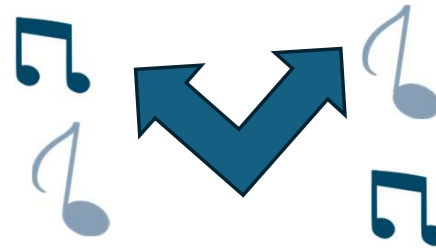
AttributeError

NameError

Console Triggers



@safe
@safe
@safe
@safe



Graphical User Interface Triggers

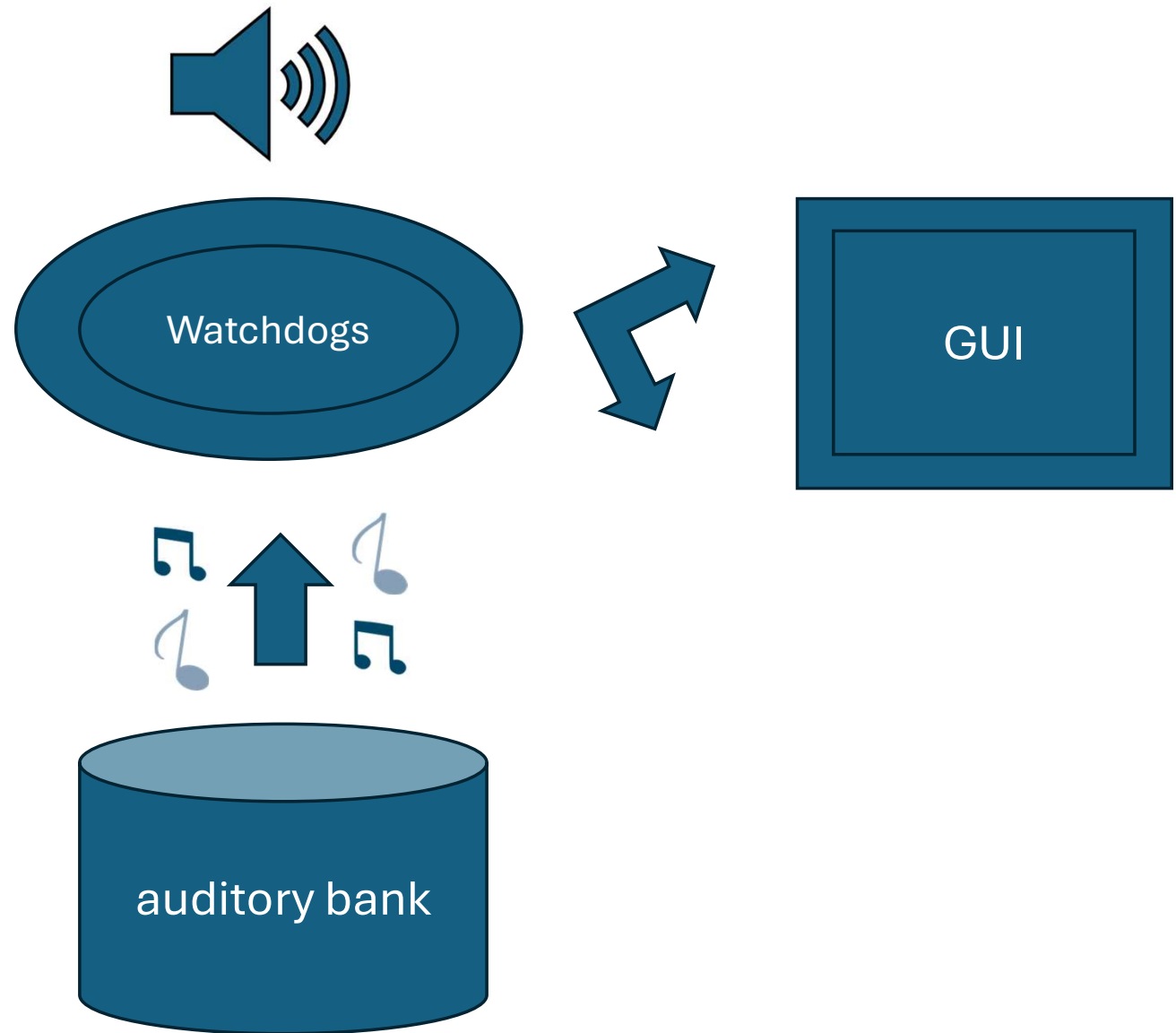


Graphical User Interface Triggers

**Widget
Lost**

**UI
Freeze**

Graphical User Interface Triggers



How is Widget Lost detected?

The programmer wrote a function to manipulate an item, but it fails, maybe the item has been removed

circle1

circle2

Color **circle1** red



How is Widget Lost detected?

The programmer wrote a function to manipulate an item, but it fails, maybe the item has been removed



Error!

Color **circle2** red



UI Freeze ❄️



How is a UI freeze detected?

A UI freeze is detected when the time since the last UI tick exceeds a threshold, meaning the thread stopped responding.

To detect the freeze, we need to set a global tick tracker

We track the time every 100 ms using a repeating heartbeat callback.

We define a non-response threshold of 2 seconds.

The elapsed time is computed as:

$$\Delta t = t_{now} - t_{last_tick}$$

software engineering freeze detection heuristic:

$$freeze \Leftrightarrow (time_now - last_ui_tick) > threshold$$

```
global last_ui_tick
last_ui_tick = time.time()
```

```
def heartbeat():
    last_ui_tick = time.time()
    root.after(100, heartbeat)
```

```
threshold = 2
```

```
if time.time() - _last_ui_tick > threshold:
    play("UI_FREEZE")
    last_ui_tick = time.time()
```